Q4 2024 Earnings Call

Company Participants

- Brice Hill, Senior Vice President, Chief Financial Officer and Global Information Services
- Gary E. Dickerson, President and Chief Executive Officer
- Liz Morali, Vice President, Investor Relations

Other Participants

- Atif Malik, Citigroup
- Brian Chin, Stifel
- CJ Muse, Cantor Fitzgerald
- Harlan Sur, J.P. Morgan
- Joe Quatrochi, Wells Fargo
- Joseph Moore, Morgan Stanley
- Krish Sankar, TD Cowen
- Srini Pajjuri, Raymond James
- Stacy Rasgon, Bernstein Research
- Tim Arcuri, UBS Securities
- Toshiya Hari, Goldman Sachs
- Vijay Rakesh, Mizuho Securities
- Vivek Arya, Bank of America

Presentation

Operator

Welcome to the Applied Materials Fourth Quarter Fiscal 2024 Earnings Conference Call. During the prepared remarks, all participants will be in a listen-only mode. Afterwards, there will be a question-and-answer session.

I would now like to turn the call over to Liz Morali, Vice President of Investor Relations. Liz, you may begin.

Liz Morali {BIO 19508304 <GO>}

Thank you. Good afternoon, and thank you for joining us for today's call. With me today are Gary Dickerson, President and CEO; and Brice Hill, CFO.

Before we continue, let me remind you that today's discussion contains forward-looking statements within the meaning of the federal securities laws, including predictions, estimates, projections or other statements about future events. Actual results may differ materially from those mentioned in these forward-looking statements as a result of risks and uncertainties. Information concerning these risks and uncertainties is discussed in our most recent Form 10-Q and 8-K filings with the SEC. We do not intend to update any forward-looking statements.

During today's call, we will also reference non-GAAP financial measures. Reconciliations of GAAP to non-GAAP results can be found in today's earnings press release and in our quarterly earnings materials, which are available on our Investor Relations website at ir.appliedmaterials.com.

I'll now turn the call over to Gary.

Gary E. Dickerson {BIO 2135669 <GO>}

Thanks, Liz. With record revenue and earnings in our fourth quarter, Applied Materials delivered a strong finish to fiscal 2024 in our fifth consecutive year of growth. I would like to recognize the hard work and commitment of our global team for delivering these outstanding results.

As this is our year end call, I'll begin by highlighting our key accomplishments over the past 12 months. A year ago in our November 2023 call, I said the company's priorities for 2024 were driving R&D programs to further differentiate our unique and connected portfolio to extend our leadership at key inflections that enable future industry growth, making operational and supply chain improvements to better serve customers, and drive productivity across the enterprise, and ensuring that we scale the company in ways that are sustainable and environmentally responsible.

Over the past year, we made significant progress in all these areas. We strengthened our position at major inflections in logic, DRAM, and advanced packaging. We delivered double-digit growth in our parts and services business. We made improvements to our operations and supply chain that supported strong cash flow and margin performance, and our key strategic initiatives are on track, including the build-out of our EPIC collaborative R&D platform and the deployment of our net-zero playbook.

In my prepared remarks today, I'll talk about three key topics. First, how large-scale secular trends are driving growth and innovation in semiconductors and why energy efficient computing is emerging as a unifying driving force for the industry. Second, how the major device architecture inflections that make up the semiconductor industry's roadmap are increasingly enabled by innovations in materials science and materials engineering, where Applied has clear leadership.

And third, as the industry roadmap becomes increasingly complex, how we are creating incremental growth opportunities for the company to offer new solutions with our unique and connected portfolio, our high-velocity collaborative R&D platform, and our advanced service products.

In the coming years, we're going to experience the biggest technology changes of our lifetimes with major advances in automation and robotics, electric and autonomous transportation, clean energy, and artificial intelligence. All of these tectonic shifts are made possible by semiconductors. And this provides a catalyst for the semiconductor industry to create and capture more value than ever before.

The biggest tectonic shift is AI, which has virtually endless applications, and therefore, the potential to transform almost every area of the economy. Deploying AI at large scale will require AI computing to be significantly more energy efficient than it is today. To realize the full potential of AI, the leading AI companies are talking about the need to drive a 10,000x improvement in computing performance per watt over the next 15 years.

To deliver energy efficiency improvements of this magnitude, evolutionary innovation will be insufficient. Instead, we see a new technology roadmap emerging made up of multiple device architecture inflections in logic, memory, and advanced packaging. This creates three significant opportunities for Applied, to deliver more value to customers and extend our differentiation in the market.

First, we have built a broad, unique, and connected portfolio of highly enabling technologies that we can supply to customers as co-optimized and integrated solutions. By combining adjacent process steps such as material deposition, etch, and material modification into an integrated system, we are providing chip makers innovative and comprehensive solutions to enable their energy efficient architecture inflections. Integrated solutions account for around 30% of our Semiconductor Systems revenue, and we expect them to become an even larger part of our portfolio in the future.

Second, we are driving earlier and broader collaborations with our customers and partners to bring next generation technology to market faster. Our global EPIC platform that we will build out over the next several years is specifically designed to accelerate cycles of learning, increase mutual success rates, and improve investment efficiencies.

In the U.S., construction of the EPIC Center in Silicon Valley is well underway and on track to come online in 2026. And we will share more details about our EPIC advanced packaging strategy at a technical summit, we are hosting for R&D leaders next week in Singapore.

And our third key opportunity is in services, where we are focused on helping customers manage increasing complexity in their business as the industry scales. We are deploying our advanced service products to help them accelerate their R&D, speed up transfer of new chip technologies from lab to fab, and then optimize device performance, yield, output, and cost in high-volume manufacturing.

This is supporting double-digit growth in our parts and service business with a high percentage of these revenues coming from subscriptions in the form of long-term agreements. These subscriptions have a high renewal rate, and the average tenure of the agreements is growing. This year, we signed our first five year service agreement with multiple customers.

Overall, AGS delivered a record quarter, a record year in their 21st consecutive quarter of year-on-year growth. Across the business, we are translating opportunities into results as major device architecture inflections grow our available market, and we gain share through the technology transitions.

In 2024, the leading edge logic companies started moving the first gate-all-around nodes from their R&D pilot lines into high volume production. We generated more than $2.5 billion of revenue from these advanced nodes in the fiscal year and expect those revenues to approximately double in 2025.

Overall, the transition from a FinFET based node to a node with gate-all-around transistors and backside power distribution grows Applied's available market from around $12 billion to approximately $14 billion for every 100,000 wafer starts per month of capacity. We also expect to capture more than 50% of the process equipment spending for the gate-all-around nodes up from the mid to high 40% range for FinFET generation fabs.

In DRAM, our revenues also grew significantly in fiscal 2024, up more than 60% year-on-year. Compute memory is a critical technology for AI data centers, and DRAM makers are accelerating their capacity plans, especially in High-Bandwidth Memory, where high performance DRAM dies are stacked and connected to a logic die with advanced packaging. The dies used in High-Bandwidth Memory are much larger than standard

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DRAM, which means that more than 3x, the wafer capacity is needed to produce the same volume of chips.

On top of this, the packaging steps needed for die stacking further increase our available market. In fiscal 2024, our HBM packaging revenues grew to more than $700 million. DRAM is a great example of how our inflection-focused innovation strategy is succeeding.

By focusing on the most enabling steps for next generation technologies, Applied has increased our share of the DRAM market by around 10 points over the past decade. Future DRAM inflections will further expand our available market as next-generation 4F2 and 3D DRAM architectures are even more materials engineering intensive.

Advanced packaging is another major device architecture inflection that provides significant improvements in the performance, energy consumption, and cost of next-generation chips. We have been investing in new technology to enable advanced packaging for more than a decade, establishing strong leadership positions in microbump and Through Silicon Vias.

In fiscal 2024, our overall advanced packaging product portfolio generated close to $1.7 billion of revenue, up 3x in the last four years. We believe this business will double in size in coming years as heterogeneous integration is more widely adopted and we introduce new solutions that grow our addressable market.

Gate-all-around transistors, backside power distribution, 4F2 and 3D DRAM, advanced packaging, and next-generation power semiconductors are all examples of device architecture inflections that are enabled by materials engineering. As a result, materials engineering, which spans all the technologies needed to deposit materials, remove or shape materials, and modify the property of materials at an atomic level is growing as a percentage of overall equipment spending at advanced nodes.

Before I hand over to Brice, let me summarize. In fiscal 2024, Applied grew revenue and earnings for the fifth consecutive year. We strengthened our position at the key technology inflections that customers will ramp in volume production over the next several years. We delivered double-digit growth in parts and services and we drove operational performance improvements across Applied and our supply chain.

As we look ahead to 2025 and beyond, we see AI and energy efficient computing remaining the key driver of innovation in the semiconductor industry. In the industry's roadmap, becoming increasingly dependent on materials engineering, which grows Applied's addressable market, and provides a tailwind for us to outperform through the investment cycle.

I strongly believe that Applied Materials has the right capabilities, strategy, and partnerships at the right time, and this puts us in a great position for the future. We are delivering differentiated solutions to our customers to help them win the key device architecture inflection races.

We are strengthening R&D collaboration with customers and partners to drive innovation and commercialization velocity, and optimize mutual success rates. And we are growing our service business by helping customers manage increasing complexity as the industry scales.

Now, I'll hand over to Brice.

Brice Hill {BIO 19639921 <GO>}

Thanks, Gary, and thanks to everyone joining today's call. We had a strong fiscal 2024 delivering record revenue and earnings per share, generating healthy operating cash flow, and distributing over $5 billion to shareholders via dividends and share repurchases. I would like to thank the entire Applied Materials team for their hard work and execution, which enabled us to achieve these excellent results.

For the full fiscal year, net sales were $27.2 billion, up 2.5% on a year-over-year basis with growth in all three business segments. Non-GAAP gross margin was 47.6%, up 80 basis points year-over-year and our highest annual gross margin rate since fiscal 2000 as we optimized our operations and made progress on value based pricing. On a year-over-year basis, non-GAAP operating profit grew 2.7% and non-GAAP operating margin was up 10 basis points. Non-GAAP earnings per share grew 7.5% year-over-year to $8.65.

For fiscal Q4, net sales were $7.05 billion, up nearly 5% on a year-over-year basis driven by solid growth in semiconductor systems and services. China declined to 30% of revenue in line with our previously communicated expectation and our historical average. Non-GAAP gross margin was 47.5%, up slightly on both a year-over-year and quarter-over-quarter basis driven by a favorable mix and operational improvements offsetting headwinds related to the lower China revenue.

Non-GAAP operating expenses were $1.28 billion or 18.2% of revenue and roughly in line with our expectations as we prioritized funding long-term strategic programs. Non-GAAP earnings per share was a record $2.32, up 9% year-over-year and benefiting from higher gross margin, higher interest income, a lower effective tax rate, and share repurchases.

Turning to the segments, semiconductor system sales were $5.18 billion for Q4, up 6% year-over-year driven by leading edge foundry-logic demand. Non-GAAP operating margin was 35.4%, down 50 basis points year-over-year given the normalizing China mix. DRAM sales declined 10% year-over-year given the elevated purchases from China in Q4 of fiscal 2023.

NAND sales were flat year-over-year. Foundry-logic sales increased 12% year-over-year fueled by robust growth at the leading edge, including increasing investments for gate-all-around nodes as customers invested to enable critical technology inflections. Sales for the ICAPS nodes, which serve customers across the IoT, communications, auto, power, and sensor markets were down year-over-year, given high demand in the year ago period.

Moving to Applied Global Services, AGS delivered record revenue of $1.64 billion in Q4, up 11% on a year-over-year basis and driven by robust growth in services, partially offset by a decline in 200 millimeter equipment sales. Non-GAAP operating margin of 30% was up 2.7 percentage points year-over-year, and non-GAAP operating income was a record $492 million.

Year-over-year, we saw increases across many operational metrics including a 7% increase in the installed base and a 10% increase in tools under service agreements. Our average contract length increased to 2.9 years, and we maintained a renewal rate of greater than 90%.

Lastly, our Display business generated revenue of $211 million in line with our expectations, as the industry experienced lower investment levels amidst ongoing weakness in end market demand. Over time, we expect there to be an increase in capital investments to support the adoption of OLED technology and IT devices like notebooks, PCs, and tablets. We are well-positioned to enable customers for the coming OLED IT inflection with our technology.

Moving to the balance sheet and cash flows, we ended the quarter with cash and cash equivalents of $8 billion and debt of $6.3 billion. Cash from operations in the quarter was $2.6 billion, capital expenditures were $407 million, and free cash flow was $2.2 billion. In total, we generated $8.7 billion in operating cash flow and $7.5 billion in free cash flow in fiscal 2024.

We distributed $1.8 billion to shareholders in the quarter including $329 million in dividends and $1.4 billion in share repurchases. For the full year, we distributed $5 billion to shareholders of which $3.8 billion was through share repurchases, up 75% from $2.2 billion in fiscal 2023. As of the end of the quarter approximately $8.9 billion remains available under our share repurchase authorization.

As we contemplate fiscal Q1, we are seeing strong demand in leading edge logic and the ICAPS nodes and sequential growth in memory. With that in mind, let me share our outlook for fiscal Q1. We expect total revenue of $7.15 billion plus or minus $400 million, and non-GAAP EPS of $2.29 plus or minus $0.18 both representing an increase of approximately 7% on a year-over-year basis.

We expect Semiconductor Systems revenue of approximately $5.3 billion, which is up 8% year-over-year, AGS revenue of approximately $1.65 billion, which is up 12% year-over-year, and Display revenue of approximately $175 million. We expect non-GAAP gross margin of approximately 48.4% driven by a favorable mix and cost and pricing improvements, and non-GAAP operating expenses of approximately $1.33 billion. We are modeling a tax rate of approximately 14%, and our outlook is consistent with trade rules currently in effect.

In closing, we had a strong fiscal 2024, with momentum across the majority of our markets, fueling record revenue and earnings per share. Our portfolio positions us to uniquely capitalize on the secular megatrends shaping the technology landscape from data center and AI to edge computing, the Internet of Things, and display. Underpinning this is our strong investment grade balance sheet, solid cash generation, and healthy shareholder distributions.

Operator, we are now ready to begin the Q&A session, please.

Questions And Answers

Operator

(Question And Answer)

Thank you. We will now open the line for questions. (Operator Instructions) And our first question comes from the line of Stacy Rasgon from Bernstein Research. Your question, please.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Hi, guys. Thanks for taking my questions. My first one, I wanted to ask what you're seeing or expecting for China mix as we go through next year. And I guess to that end, I was a little surprised you suggested you saw ICAPS strength going into Q1. Do you think -- I guess maybe you can talk to us about how that ICAPS strength in the near-term splits out between China and the rest of the world? And how you see China mix either sustaining at 30% or moving off that number as we go through next year?

A - Brice Hill {BIO 19639921 <GO>}

Great. Thanks for the question, Stacey, and hello. Yes, the China mix, a couple signals there. So 30% in Q4, as we reported, and that's normalized and come down after we served that China DRAM demand that we had in earlier quarters. Our outlook in Q1 also contains approximately 30% for the China mix.

When we think about ICAPS across the world, it is still very healthy. So we said, it's lower than our prior Q4, so Q4 of '24 was a little bit lower than Q4 of '23, but the ICAPS market is still very healthy and it's healthy globally. It's healthy in China and it's healthy across the world from a total level perspective.

Having said that, we expect the ICAPS markets to grow over time. We've talked about mid to high single digits at the device level. We expect customers to continue to add capacity. There are signs to watch for. There are slower end markets in ICAPS. Automotive, industrial, analog, image sensors are all markets that have been slower. So we've kind of been looking to see, if the investment rate would go a little bit lower, but as far as our Q4 and our Q1, it remains strong.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Got it. Thanks. And for my follow-up then, if China mix isn't going up, gross margins are, so it doesn't sound like the gross margin increase is due to an increase in China mix. So can you talk about the drivers of that gross margin increase into Q1? And I guess maybe talk to sustainability as we go through next year? Do you think you'd get back closer to the target ranges for gross margins as we go through the end of the year?

A - Brice Hill {BIO 19639921 <GO>}

Yes. Thanks for the question on that. So, our gross margin, the way we want to shape that from your modeling perspective and where the business is, we think our underlying rate has improved to about 48.0% at this point.

So it has improved, and we've made improvements top to bottom in the business, logistics, our inventory management, scrap management, cost overall, and our implementation of value pricing. So, a lot of elements working on the gross margin equation. In Q1 in particular, we're guiding 48.4%. So above that 48.0% sort of baseline, and that's based on really strong product mix that we have in Q1. So, we think 48% is the right level for you to think about longer term, and we'll of course continue to work on improving that.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes. Stacy, this is Gary. One of the things also that we're very focused on is winning the inflections. There's a lot of really great inflections that provide a tailwind for Applied. And we're bringing enabling technology to our customers in a number of different markets.

So one thing that is also a tailwind for us is pricing improvement as we're shipping more valuable products. So that's part of what you're seeing in that increased gross margin, and I think that we have a great opportunity to continue to drive that going forward.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Got it. Thank you, guys.

Operator

Thank you. And our next question comes from the line of CJ Muse from Cantor. Your question, please.

Q - CJ Muse {BIO 6507553 <GO>}

Yes. Good afternoon. Thank you for taking the question. I know you guys don't want to talk about WFE, but I was hoping you could speak to it directionally. How are you thinking about '24, '25, more importantly, growth? And maybe perhaps more importantly in terms of the change in the administration, how are you thinking about the potential risk in terms of added restrictions or perhaps going the other direction? We'd love to hear your first thoughts there.

A - Brice Hill {BIO 19639921 <GO>}

Okay. Thanks for the question, Stacy. Sorry, CJ, and thanks for staying up for us. So on the WFE, directionally for '24, so we just closed our fifth year of growth. So, we'll see what the print is across the whole industry, but we would expect it to have shown growth. And then for '25, if you zoom out, we're talking about $1 trillion semiconductor industry by 2030. I think that's consistent across most of the industry having that view.

And we certainly have that view given the added wafer starts and the added capacity across the industry, basically every single year. So it's hard to guarantee '25, but I'll just call out that our Q4 was growth year-over-year, our Q1 is growth year-over-year, and we just completed those five years. And we're starting to see that leading edge component of logic accelerate as the gate-all-around nodes are being invested in across the world.

Last, as far as the change in administration, it's early, we really can't speculate on what might change, so we'll have to wait for more input on that one. Thank you.

Q - CJ Muse {BIO 6507553 <GO>}

Very helpful. As a follow-up on gross margins, the 48% now as a floor is a fairly pretty good accomplishment. So, we'd love to hear kind of how you got there? And I guess, how to think about growth beyond that over time.

A - Brice Hill {BIO 19639921 <GO>}

Yes. I don't think I can be as aggressive to call a floor, but I won't call a peak either. So, 48.4% for Q1 that's mixed positive for Q1, but we do expect to continue to make improvements on that 48.0% baseline.

So, we continue to have significant cost projects in our pipeline. And as we introduce new equipment and improve the value of the equipment that is existing, we'll improve our value pricing. So, we expect to continue to make progress.

Q - CJ Muse {BIO 6507553 <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of Vivek Arya from Bank of America. Your question, please.

Q - Vivek Arya {BIO 6781604 <GO>}

Thanks for taking my question. On the leading edge, the three big kind of foundries and the idea among them, one of them is doing extremely well, the other two not as much. So, when you say that leading edge is very strong, is the expectation that the one player that is doing very well continues to stay very strong.

And the others, even if they fall behind, are still sufficient to give the overall industry a strong growth outlook for next year. So I'm just -- I guess, the direct question is, how much have you handicapped this kind of dichotomy in the market, where one player is doing extremely well, and the other two are not doing as well?

A - Brice Hill {BIO 19639921 <GO>}

Thanks for the question, Vivek. The way that we think about forecasting demand long term for the business really starts with the end markets. Our view, if you think about data center, PCs, smartphones, all the things that artificial intelligence that drive the leading edge. We think the eventual footprint of capacity will need to match those end market requirements.

And it will be sort of independent of which foundries are serving what amount of that demand. And so at a high level, we haven't really changed our expectations from the amount of leading edge capacity that needs to be installed. I would say as customers change their schedules and their microfactory forecasts, our forecast really hasn't changed.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Vivek, this is Gary. What I would add is that I've been spending a lot of time with all of these different companies. The really big focus for everybody is improvement in energy efficient computing AI is driving a significant amount of growth, those die sizes are very large. And so when we look forward, not just in '25, but we look over longer term, we see significant growth in leading edge foundry-logic.

And the good thing for Applied is that those improvements in energy efficient computing are really enabled through architecture changes like gate-all-around or backside power distribution, where in every one of these architecture inflections, you're improving the power efficiency 20% to 30%. So it's really, really important. One of the customers I was talking to said that every percent of energy efficiency improvement matters.

And as we've talked about, we're really well positioned to gain share as our customers are going through those inflections with new gate-all-around nodes and backside power. So, that gives us a really great tailwind going forward. As Brice said, we really look at this from an overall market standpoint. We're pretty optimistic relative to the growth in leading edge foundry logic.

A - Liz Morali {BIO 19508304 <GO>}

Thanks, Vivek. We'll take our next question.

Operator

Thank you. And our next question comes to the line of Atif Malik from Citigroup. Your question, please.

Q - Atif Malik {BIO 15866921 <GO>}

Hi. Thank you for taking my question. I have a question for Brice. Brice, I believe on the last earnings call, you talked about China being 20% of your sales on a normalized basis next year, is that still your view? And some of your peers have talked about a decline anywhere ranging from mid-teens to down 30% for their China sales. And I know you're not talking about WFE next year, but is that the right decline for China sales for you guys as well between 15% and 30%?

A - Brice Hill {BIO 19639921 <GO>}

Hi, Atif. We have some shaking heads in the room here. We don't recognize, if I heard you right, we don't recognize the 20%. But I think the last quarter was 32%, this quarter reported Q4 is 30% for our China as a share of total revenue.

And our outlook quarter Q1 is approximately 30% also, so we think that's a normalized rate. It was elevated for a few quarters, even up into the mid-40%s. And that was because we were shipping that DRAM demand to some specific China customers, when it was allowed. Since then, it's come back down to the 30% range.

And as you've highlighted in our outlook, that's consistent with our Q1. And we think that's a -- when we look over the past several years, we think 30% is approximately normal for the company. And that does include all of our businesses. That includes the semi systems, that includes our services business, and our display business, which has significant sales in China.

Q - Atif Malik {BIO 15866921 <GO>}

Thank you for the clarification.

A - Brice Hill {BIO 19639921 <GO>}

Yes.

Operator

Thank you. And our next question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi. Good afternoon. Thank you so much for taking the question. Gary and Brice, I think you both mentioned the concept of value based pricing, a couple of times in your script and how that's driving gross margin. I know this isn't necessarily a new initiative at Applied, but if you can kind of expand on what you're doing exactly with your customers, is it primarily tied to your IMS offering or is it broader than that? And which inning are you in as it pertains to you guys essentially trying to capture value? Thank you.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes. Thanks, Toshiya. I would say third inning. And the reason I say that is because we are, since we had the COVID events and we had supply chain and we had changes in our cost, we also had to reconcile with that. We've always had value pricing in the company. We've always thought about the value of the tools, but that was a bit of a shock in terms of what the cost levels were coming through the supply chain.

So, we really had to stimulate our evaluation of the value being provided for each application, and especially those integrated applications that are unique to the company. And so, we're in the process of strengthening our training process and our analytical process and our communications with the customers. And it's since the tools that we're providing and the systems we're providing, we do think create more value for the customers. We're making progress in that.

A - Brice Hill {BIO 19639921 <GO>}

Yes. Toshiya, if I look across all of the different businesses within Applied, I've never been more optimistic regarding the technology pipeline and product pipeline that we have in our positions around these key inflections that are important for energy efficient computing. So as you mentioned, the integrated platforms, that's about 30% of our revenue. And that's -- those are absolutely crucial for these technology inflections.

But I would say that if I look at foundry, logic or DRAM, where we've gained 10 points of share over the last 10 years or advanced packaging, or even in ICAPS, we have new products in the pipeline that will expand our available market and are incredibly valuable for some of these segments. I think, we have a really robust pipeline of capabilities, and that'll help us from a value pricing standpoint. So as Brice said, I think, we have a lot of room to continue to grow.

Q - Toshiya Hari {BIO 6770302 <GO>}

Great. Thank you.

Operator

Thank you. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

Q - Tim Arcuri {BIO 3824613 <GO>}

Thanks a lot. Brice, you gave a number just now. You said that advanced node revenue is going to double. You said it was $2.5 billion, I think this year. And you said it's going to double next year. Not all that's going to be incremental though, because some is going to cannibalize as I would think, N3, N4, N5. So, do you have a sense of how much of that would be incremental? I mean, I'm sure that some of it is, but do you have a sense of how much of it would be?

A - Brice Hill {BIO 19639921 <GO>}

Thanks, Tim. So yes, this approximate doubling has to do with our shipments to the gate-all-around nodes across the industry. What we've said is in the fiscal '24, we shipped approximately $2.5 billion of equipment to gate-all-around nodes, and we expect that to approximately double next year.

And if I think about it, we've also described the increment for gate-all-around for us as approximately $1 billion raising what was a $6 billion investment for 100,000 wafer starts of capacity, wafer starts per month to $7 billion. So, the increment would be about 1/6, if you're thinking about it from that perspective. And then, I think most of our shipments at this point are on the, in the leading-edge logic are towards gate-all-around nodes. So, most of that is replacing revenue that was at prior nodes in the past.

Q - Tim Arcuri {BIO 3824613 <GO>}

Very, very helpful. Thank you, Brice. And then, Gary, I know you talk about $1 trillion in semiconductor revenue, that's this much value longer-term number. But my question is, how much WFE spending do you think has to be spent to support that much revenue?

And really, it kind of gets to what the longer-term WFE intensity is, which used to be 13.5% before COVID, and it got to like 17% in 2023, but it's been coming down since then, as China's come down. So how do you think about, like for a $1 trillion worth of semiconductor revenue, how much WFE do you think the industry has to spend to support that? I don't know, Brice, if you want to answer that, or Gary.

A - Brice Hill {BIO 19639921 <GO>}

Yes, I can jump in on that, because we have seen elevated intensity, the WFE investment, the equipment investment over semiconductor revenues. We've seen that intensity increase largely on that significant ramp in China over the past couple of years, as China ramped up capacity.

So, we do expect it to decrease a little bit over time as we look forward in the forecast. But we're comfortable at this point that it should stay in the mid-teens, somewhere in the mid-teens as a metric.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes, Tim. The other thing I would add is that we're very deeply engaged with all of our customers, I would say more so than ever. This concept of high-velocity co-innovation, where we're co-innovating with our customers a decade out over multiple technology nodes, has never been stronger with all of our innovative technologies, our device integration teams, really deep partnerships with customers.

And what we see going forward, besides overall capital intensity is increasing intensity in materials engineering and the technologies from Applied Materials. So, I think, again, we have very, very deep visibility across all of these different market segments, and that will put us in a good position.

Q - Tim Arcuri {BIO 3824613 <GO>}

Thank you, both.

Operator

Thank you. And our next question comes from the line of Krish Sankar from TD Cowen. Your question, please.

Q - Krish Sankar {BIO 16151788 <GO>}

Yes. Hi. Thanks for taking my question. Gary or Brice, I just wanted to follow up, again on the value based pricing argument. Is part of it driven by some of these technologies like gate-all-around, where it's really difficult to do some of the vertical EPI steps, and therefore, there's a lot more value added to it.

The reason I'm asking is once you get past gate-all-around, go to backside power delivery or even 6F2 DRAM, does that intensity come down? In other words, gate-all-around is so difficult to do, the next steps might not be as critical. I'm just kind of curious how to think about this value pricing, and whether it was a gate-all-around specific thing.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Hi, Krish. This is Gary. Thanks for the question. So, I would say that the entire industry is focused on energy efficient computing. And we've heard some of the system companies talking about 100x improvements in five years or 10,000x improvements in 15 years.

It's really hard to accomplish those types of improvements. And the value why do these companies go to the most advanced nodes? They go there even though those wafers are more expensive, because one CEO was telling me even 1% improvement in energy efficient computing is worth a lot. So, when you think about all of these different nodes, yes, gate-all-around is difficult, backside power is difficult, 4F2 and 3D DRAM are difficult. All of these areas are very, very challenging.

And we have unique capability with our integrated platforms. We have one platform with multiple technologies that enables a 50% improvement in resistance. And obviously, that's really critical for energy efficient computing. So, I don't think this is a one-shot thing with the first generation of gate-all-around. All of these things, I think, are critical for customers and Applied extremely well positioned.

Q - Krish Sankar {BIO 16151788 <GO>}

Thanks, Gary. Very helpful.

Operator

Thank you. And our next question comes from the line of Harlan Sur from J.P. Morgan. Your question, please.

Q - Harlan Sur {BIO 6539622 <GO>}

Good afternoon. Thanks for taking my question. On the team's services business, strong performance, 21 consecutive quarters of year-over-year growth, and that trend clearly looks to sustain going into an improving semiconductor demand environment next year.

Additionally, you guys have driven a much more annuity like recurring and attractive revenue profile. However, operating margins are still about 100 basis points below the 31% level that you drove for five, six consecutive quarters in fiscal '21 and '22 time frame, and that was at a lower revenue base.

And then maybe about 200 basis points, 300 basis points below your prior long-term targets for low 30% operating profitability on revenue targets that were also lower. So, what's been responsible for the lower operating margin expansion? Is there still line of sight to getting into the low 30% range on continued AGS growth or is there something sort of structurally that has changed?

A - Brice Hill {BIO 19639921 <GO>}

Yes. Harlan, thanks for the question. There is something structurally that's changed. We have begun allocating more of our corporate expenses and our central expenses that are supporting each of our segments during this year, and that was accountable for a portion of the reduction in operating profit there.

So I think, you see us improving our operating profit for that business over the last eight quarters, generally. And I think the expectation is we'll continue to be able to make improvements even with those allocations, that allocation increase that we made.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes, Harlan. Also there's tremendous pull as customers are going through these inflections in all of these different markets to accelerate time to market for those innovations, and then transferring technology, first-time right, faster technology transfer, and then ramp into high-volume manufacturing, those are big opportunities.

We're also driving tremendous innovation in our services, and I've been meeting with a lot of the different customers here over the next quarter, there's real traction for many of these service innovations. So that will continue to drive the value we create for our customers and drive our margins higher in the future.

Q - Harlan Sur {BIO 6539622 <GO>}

No. I appreciate that. And maybe just a quick follow-up. So as you guys focus on more value capture, especially with your integrated solutions, Gary, you spend a lot of time talking about sort of new technology enablement with these integrated systems. Do your customers also get the additional benefit in terms of better operational and manufacturability dynamics?

I mean, I can imagine with an integrated system, you eliminate a lot of the wafer transport, transfer activities, which ultimately increases process module throughput, increases cycle time, you also potentially get smaller tool footprint, right? Is this an accurate assessment in any way to kind of quantify some of these operational benefits with your integrated systems?

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes. Thanks for the question. So, we're always focused on improvements in energy efficient computing and all of these different architectures, but there is also a huge focus in cost innovation. And cost innovation when we're combining these different technologies together is one aspect.

But the other thing is really helping our customers optimize these new device architectures. So, when we're in these technology discussions with a number of different companies, there are cases, where we can simplify processes that really enable them to drive lower costs. Another example is with our pattern shaping technology that we've talked about, where our customers can reduce the number of EUV steps. So there are a number of those kinds of opportunities.

And then getting back to services. Again I was with one of our biggest customers here recently, and we were talking about service innovation that can drive their overall operating costs lower. So there -- this is also a really big focus and I think an opportunity both for our systems business and our service business.

Q - Harlan Sur {BIO 6539622 <GO>}

Thank you, Gary.

Operator

Thank you and our next question comes from the line of Joe Quatrochi from Wells Fargo. Your question, please.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Hey, thanks for taking the question. You mentioned in the prepared remarks that you were seeing DRAM customers accelerate capacity plans especially for HBM. I think most people expect HBM to continue to be strong spending, but are you seeing customers look to add capacity and for conventional DRAM? I just want to make sure, I understand that comment right. And then how do we think about just that the growth that you talked about sequentially for memory into the January quarter?

A - Brice Hill {BIO 19639921 <GO>}

Hi, Joe. We do see customers adding capacity in DRAM in total more way for start capacity. I actually haven't calculated how much of that was driven by HBM. What we would say is about 10% of DRAM wafers right now are allocated towards HBM production, High-Bandwidth Memory and the High-Bandwidth Memory demand is growing at about a 30% rate.

So we do see more capacity allocated each quarter to High-Bandwidth Memory, and we do see the growth rate there very high. My sense is that doesn't equate to exactly what's being added in the overall capacity footprint. But anyway we do see capacity increases in the in the DRAM customers, and we do see that market continue to be fairly strong.

Operator

Thank you. And our next question comes from the line of Srini Pajjuri from Raymond James. Your question, please.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Thank you. I have a follow-up to the previous question, Brice. If I look at your DRAM, obviously very strong in fiscal '24. I think, you said 60 plus percent growth. And you kind of quantified the HBM at $700 million roughly. I think that's roughly about 20 points of growth.

So as I look out to the next, I guess fiscal year, I'm just trying to understand how to think about overall DRAM growth, because HBM will obviously grow. And I believe, there was some China in last 12 months that's probably declining or already declined. So all in do you expect I guess DRAM to grow next year or do you think it's going to be a challenge? Thank you.

A - Brice Hill {BIO 19639921 <GO>}

Yes. So, a couple comments there, long-term we've talked about the $1 trillion market. And our view that $1 trillion semiconductor market in our view that capacity will be continued to be added across the ecosystem on the wafer side to support that. I think, we've articulated in the past that for WFE, we expect about 2/3 to be foundry-logic and 1/3 to be memory. We haven't gotten specific about what the share is for DRAM.

But right now, today, in our Q4 and our Q1 guide, we're seeing a strong DRAM market, and that's a continuation. And we talked about the pull of High-Bandwidth Memory from the artificial intelligence usage models. So I think, it's fair that we expect capacity to be added over time, and we're not making a call on '25, but those are the dynamics.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Got it. And then maybe another quick one on NAND. I know you guys have been fairly cautious on WFE spending, when it comes to NAND, but there is definitely some optimism about tech transitions and in particular in a MOLY-B [ph] transitions. So, just want to hear your thoughts about how you're thinking about NAND for next year in terms of tech transitions and the opportunity for you guys.

A - Brice Hill {BIO 19639921 <GO>}

Yes. From a macro perspective, similar to the comments on DRAM, we do see a little bit of growth in NAND in our Q1 outlook, and the dynamic is slightly different at the macro level for NAND than DRAM. On the DRAM side, we do see added capacity, more wafer starts across the ecosystem.

On the NAND, we haven't seen that, because I think the bit rate density, so the actual shrink rate of the NAND technologies has been so impressive and so high that it's actually delivered the amount of bits that are in the demand profile. And so, there hasn't been new wafer starts needed on the NAND side, so most of that has been an upgrade market. So that dynamic is slightly different. But in any case, we see a little bit of an uptick in the NAND in our outlook for Q1.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of Brian Chin from Stifel. Your question, please.

Q - Brian Chin {BIO 5763426 <GO>}

Hi there. I appreciate the time this afternoon. It sounds like you still view 30% as a normalized level of China sales. But of course, it did go higher than normalized into the 40s in recent quarters. So I was curious, is it reasonable that China could go below 30% in

'25 if areas like leading edge foundry or DRAM improve and ICAPS remains more stable, plus or minus?

A - Brice Hill {BIO 19639921 <GO>}

Thanks for the question, Brian. I think that most of the variability for us will be based on the health of the ICAPS market in China, that's the vast majority of our business right now. We have served most of the DRAM and NAND business that we can serve, and we can't serve leading edge in China. So most of the vast majority of the business is ICAPS, and it'll just depend on how that ICAPS market evolves.

Earlier, I shared that there's a couple of signals that say the rate of investment might be slower. Those are higher inventory positions and slow end markets that are automotive, industrial, image sensors, and analog are markets that are slower. But having said that, our Q4 and Q1 outlook is still strong for ICAPS across the world. And so we'll just have to see how that matures. We do expect customers to continue to add capacity over the medium term.

Q - Brian Chin {BIO 5763426 <GO>}

All right. Thank you. Maybe if I could just ask a little bit of a longer term question here. ASML earlier today was upbeat on its outlook for AI driven DRAM spinning overall. They also talked about lithography intensity increasing for DRAM toward decade end. This seems to differ, I guess, a little bit from your prior messaging around the potential timing and introduction of technologies like vertical transistor DRAM and even 3D DRAM. Can you maybe compare and contrast, where the differences here might arise?

A - Brice Hill {BIO 19639921 <GO>}

I'll say one thing, and then, Gary will help here. I think, my view is it's complementary, very complementary. So as you know, there's more business and more demand on the Litho side of the equation that of course requires more equipment to do the actual materials. So Gary has a comment.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes, we're deeply engaged with all of the memory companies on their DRAM roadmaps. So relative to the vertical channel transistor, also known as 4F2, I think that's where everybody's focused in the near-term. We believe that inflection will be more materials engineering intensive.

And again, we're really deeply engaged on the key technologies that will enable that architecture inflection. And I would say that everyone is also focused on 3D DRAM and that one is further out in time and that is significantly more materials engineering intensive.

So again, I can't really comment on what other people are seeing, but I would say that we're deeply engaged with all of those different companies on those architectures. We're working with their integration teams, their device teams, and we have high confidence that materials engineering intensity will increase with those inflections.

Q - Brian Chin {BIO 5763426 <GO>}

All right. Thank you.

Operator

Thank you. And our next question comes from the line of Joseph Moore from Morgan Stanley. Your question, please.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you. I wanted to follow up the strength in ICAPS. You sort of said there's weakness in automotive, industrial, analog, and image sensors, that's a lot of the trailing edge category. So I'm wondering, what is that leave that's doing well. And is that sort of imply that it's geopolitically related spending? Is there just what aspect of it is it that's still strong?

A - Brice Hill {BIO 19639921 <GO>}

Yes. Thanks, Joe. I think power, some of the power related components, and I think some of the microcontrollers are stronger than those markets. Those are two that I recall from the discussions. So I think, what we see across ICAPS is it's continued to be fairly strong. It wasn't quite as high in our Q4 '24 as in our Q4 '23, but still a very strong market. So, we'll just have to watch how this evolves.

Our perspective is that customers will continue to add capacity as the underlying market continues to grow towards this $1 trillion 2030 semiconductor market, but utilizations right now are a little bit lower than the ideal utilization level. So, we do think there could be some slowing of investment as we go through this year.

A - Liz Morali {BIO 19508304 <GO>}

Thanks, Joe. We have time for one last question.

Operator

Certainly. And our final question for today then comes from the line of Vijay Rakesh from Mizuho. Your question, please.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Yes. Hi. Just a quick question on the China side. Gary and Brice, when you look at the China revenues, do you expect it to kind of hold at these levels going through calendar '25?

A - Brice Hill {BIO 19639921 <GO>}

Thanks, Vijay. In our Q1 guide, we are highlighting that it is also approximately 30%. So -- and looking historically, that's been the number that we're comfortable, that we've seen historically. And I'll just say that the market for us, I highlighted a few minutes ago, we're not serving much DRAM or NAND, and no leading edge at this point, it's mostly the ICAPS business.

And so the fact that it's been restricted to the ICAPS business at this point, it'll really change over time with the strength of that particular end market. We do expect that to grow over time. We'll have to see how '25 shakes out as you know, as the market matures. Thanks for the question.

Operator

Thank you. This does conclude the question-and-answer session of today's program. I'd like to hand the program back to Brice for any further remarks.

A - Brice Hill {BIO 19639921 <GO>}

Thank you, operator. I'm pleased with the strong performance we delivered in Q4 and for our fiscal year. We're in a strong position with an industry-leading portfolio of products and services, and poised to uniquely benefit from the secular megatrends driving technology demand. Thank you for joining today's call.

And Liz, please close the call.

A - Liz Morali {BIO 19508304 <GO>}

Thank you, Brice, and thanks to everyone for joining the call today. A replay of today's call will be available on the Investor Relations website by 5:00 p.m. Pacific Time today. Thank you for your continued interest in Applied Materials.

Operator

Thank you, ladies and gentlemen for your participation in today's conference. This does conclude the program. You may now disconnect. Good day.
